

LOW STYRENE SBR PATENT LANDSCAPE AND SYNTHESIS REPORT

1. ABSTRACT

This report reviews two primary patents relevant to low styrene styrene-butadiene rubber (SBR) synthesis and composition control. The target is defined as SBR materials with reduced styrene incorporation, typically achieved by controlling the butadiene-to-styrene feed ratio and the polymerization route. The patent landscape spans two distinct approaches: a solution-polymerized hydroxyl-terminated SBR used as a soft segment in polyurethane elastomers, and a patent explicitly directed to preparing SBR having low combined styrene content. Together, they show that low-styrene SBR can be addressed either as a standalone polymerization objective or as a tailored intermediate for downstream elastomer systems.

US20190112458A1 discloses an anionic solution polymerization route using a dilithium initiator, cyclohexane/tetrahydrofuran solvent system, and ethylene oxide end-capping to produce hydroxyl-terminated SBR. The examples vary the butadiene-to-styrene ratio, active lithium ratio, temperature, and reaction time, yielding SBR soft segments with molecular weights from 1000 to 8000 and glass-transition temperatures around -20 to -30 °C. Although the patent is framed around polyurethane elastomers, it provides explicit solution-polymerization recipes and styrene feed control that are directly relevant to low-styrene SBR design.

EP3243844A4 is more directly aligned with the target, as its title and description explicitly state a method for preparing styrene-butadiene rubber having low combined styrene content. The extracted evidence is largely bibliographic and classification-based, but it clearly identifies aqueous emulsion polymerization as the process family and low combined styrene content as the objective. Overall, the evidence suggests that low styrene SBR synthesis is primarily governed by monomer feed ratio control, polymerization route selection, and downstream end-use requirements such as tire performance, with solution and emulsion routes both represented in the patent set.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20190112458A1

- Patent Title: Solution Polymerized Styrene-Butadiene Rubber-Polyurethane Elastomer Material
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- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses an anionic solution polymerization route to make hydroxyl-terminated styrene-butadiene rubber, using a dilithium initiator prepared from isoprene and lithium metal. Butadiene and styrene are polymerized in cyclohexane with tetrahydrofuran present in the initiator preparation, followed by ethylene oxide end-capping and acid/ethanol workup to isolate the hydroxyl-terminated SBR. - Dilithium initiator concentration: 0.8 mol/L — Example 1 - Lithium sheet: 70 g — Example 1 - Tetrahydrofuran: 500 ml — Example 1 - Isoprene: 200 ml — Example 1 - Cyclohexane: 500 ml — Example 1 - Reaction temperature: about 0 °C — Example 1 - Initiator stirring time: 6 h — Example 1 - Low temperature hold: 24 h — Example 1 - Ethylene oxide: 100 ml — Example 1 - Calcium hydride: 10 g — Example 1 - Butadiene to styrene mass ratio: 100:25 — Example 1 - Monomer mass fraction: 10% — Example 1 - Active lithium to (butadiene+styrene) mass ratio: 1:280 — Example 1 - SBR synthesis temperature: 50 °C — Example 1 - SBR synthesis time: 4 h — Example 1 - Ethylene oxide to active lithium mass ratio: 100:18 — Example 1 - End-capping time: 12 h — Example 1 - Hydrochloric acid treatment time: 4 h — Example 1 - Molecular weight: 3000 — Example 1 - Glass transition temperature: -25.9 °C — Example 1 - Lithium sheet: 60 g — Example 2 - Active lithium to (butadiene+styrene) mass ratio: 1:430 — Example 2 - Ethylene oxide to active lithium mass ratio: 100:20 — Example 2 - Molecular weight: 4500 — Example 2 - Glass transition temperature: -28.5 °C — Example 2 - Lithium sheet: 90 g — Example 3 - Butadiene to styrene mass ratio: 100:30 — Example 3 - Active lithium to (butadiene+styrene) mass ratio: 1:90 — Example 3 - SBR synthesis temperature: 45 °C — Example 3 - Molecular weight: 1000 — Example 3 - Glass transition temperature: -28.1 °C — Example 3 - Lithium sheet: 55 g — Example 4 - Butadiene to styrene mass ratio: 100:35 — Example 4 - Active lithium to (butadiene+styrene) mass ratio: 1:790 — Example 4 - SBR synthesis temperature: 55 °C — Example 4 - Molecular weight: 8000 — Example 4 - Glass transition temperature: -28.3 °C — Example 4 - Soft segment to isocyanate mass ratio: 100:(15-45) — Claims - Soft segment+isocyanate to chain extender+cross-linking agent mass ratio: 100:(4-12) — Claims - Chain extender to cross-linking agent mass ratio: 100:(17-100) — Claims - Soft segment molecular weight: 1000-8000 — Abstract/Claims - Soft segment glass-transition temperature: -20 °C to -30 °C — Abstract/Claims - Reaction temperature during butadiene-styrene rubber synthesis: 45-55 °C — Claims/Description - Reaction time during butadiene-styrene rubber synthesis: 3.5-4.5 h — Claims/Description - Butadiene to styrene mass ratio: 100:(25-35) — Claims/Description - Ethylene oxide to active lithium mass ratio: 100:(10-20) — Claims/Description - Active lithium to (butadiene+styrene) mass ratio: 1:(90-790) — Claims/Description - Dilithium initiator concentration: 0.5-1.0 mol/L — Description - Prepolymer stirring temperature: 95 °C —

Example 1 - Prepolymer vacuum stirring time: 2.5 h — Example 1 - MDI amount: 4.35 g — Example 1 - Prepolymer reaction time: 3 h — Example 1 - NCO content: 7% — Example 1 - Final elastomer stirring temperature: 80 °C — Example 1 - Final elastomer vacuum stirring time: 2 h — Example 1 - BDO amount: 0.92 g — Example 1 - TMP amount: 0.16 g — Example 1 - Dropping rate: 1 s/drop — Example 1 - High-speed stirring rate: 400 rad/min — Example 1 - Curing temperature: 100 °C — Example 1 - Curing time: 20 h — Example 1 - Post-demoulding time: 7 d — Example 1 - Medium & Water Role: The core SBR synthesis is solution polymerization in cyclohexane, with tetrahydrofuran used during initiator preparation. Water appears only as an ice-water bath condition and not as the polymerization medium; later workup includes hydrochloric acid treatment and ethanol precipitation, but no aqueous emulsion polymerization is disclosed. - low styrene content: Butadiene to styrene mass ratio: 100:25; 100:30; 100:35 — Examples 1-4; Claims disclose 100:(25-35)

Medium & Water Role - The core SBR synthesis is solution polymerization in cyclohexane, with tetrahydrofuran used during initiator preparation. Water appears only as an ice-water bath condition and not as the polymerization medium; later workup includes hydrochloric acid treatment and ethanol precipitation, but no aqueous emulsion polymerization is disclosed. - Core reaction medium: Organic solvent solution polymerization (cyclohexane-based; tetrahydrofuran used in initiator preparation)

low styrene content - Butadiene to styrene mass ratio: 100:25; 100:30; 100:35 — Examples 1-4; Claims disclose 100:(25-35)

Relevant Experimental Evidence - Example 1: Produced hydroxyl-terminated SBR with butadiene:styrene = 100:25, Mw 3000, and Tg -25.9 °C, then converted it into a polyurethane elastomer. - Example 2: Increased the active-lithium ratio and obtained higher molecular weight SBR (Mw 4500) with Tg -28.5 °C. - Example 3: Used butadiene:styrene = 100:30 and obtained a lower molecular weight SBR (Mw 1000) with Tg -28.1 °C. - Example 4: Used butadiene:styrene = 100:35 and obtained the highest molecular weight among the examples (Mw 8000) with Tg -28.3 °C. - Example 1: Demonstrated polyurethane formation from the SBR soft segment using MDI, BDO, and TMP with cast molding and curing.

Technical Relevance This patent is relevant because it explicitly discloses solution-polymerized SBR recipes with controllable styrene feed ratio, which is central to low styrene content design. It also provides practical polymerization and end-capping conditions that can be adapted when targeting lower styrene incorporation in SBR-based materials.

Patent 2

- Patent Number: EP3243844A4
- Patent Title: Method for preparing styrene-butadiene rubber having low combined styrene ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent is directed to a method for preparing styrene-butadiene rubber having low combined styrene content. The extracted evidence indicates an emulsion polymerization family process in an aqueous medium, but the provided text does not include worked examples or detailed recipe conditions in the extracted block. - Publication date: 2018-10-31 — bibliographic data - Priority date: 2015-11-30 — bibliographic data - Filing date: 2016-11-25 — bibliographic data - Medium & Water Role: The extracted evidence points to aqueous emulsion polymerization as the core reaction medium. Water is therefore part of the polymerization medium, not merely a downstream workup solvent, although no detailed recipe or post-treatment water steps are provided in the extracted block. - low styrene content: low combined styrene content

Medium & Water Role - The extracted evidence points to aqueous emulsion polymerization as the core reaction medium. Water is therefore part of the polymerization medium, not merely a downstream workup solvent, although no detailed recipe or post-treatment water steps are provided in the extracted block. - Core reaction medium: Aqueous emulsion polymerization

low styrene content - low combined styrene content

Relevant Experimental Evidence - No worked examples were detected in the extracted evidence. - The title explicitly states the objective of low combined styrene content. - Classification data indicates aqueous medium and emulsion polymerisation. - The record identifies the patent family as related to SBR copolymers with styrene. - The extracted block does not provide numerical polymerization recipe details.

Technical Relevance This patent is directly relevant because it explicitly targets low combined styrene content in SBR, matching the user’s target material definition. Even though detailed examples are absent in the extracted evidence, the patent confirms an emulsion-based route for low-styrene SBR manufacture.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	low styrene content	Key Findings
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US20190112458A1	Not disclosed	2019	The core SBR synthesis is solution polymerization in cyclohexane, with tetrahydrofuran used during initiator preparation. Water appears only as an ice-water bath condition and not as the polymerization medium; later workup includes hydrochloric acid treatment and ethanol precipitation, but no aqueous emulsion polymerization is disclosed.	Butadiene to styrene mass ratio: 100:25; 100:30; 100:35 — Examples 1-4; Claims disclose 100:(25-35)	Example 1: Produced hydroxyl-terminated SBR with butadiene:styrene = 100:25, Mw 3000, and Tg -25.9 °C, then converted it
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EP3243844A4	Not disclosed	2018	The extracted evidence points to aqueous emulsion polymerization as the core reaction medium. Water is therefore part of the polymerization medium, not merely a downstream workup solvent, although no detailed recipe or post-treatment water steps are provided in the extracted block.	low combined styrene content	No worked examples were detected in the extracted evidence.
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US20190112458A1 uses solution polymerization in cyclohexane with a dilithium initiator and ethylene oxide end-capping, while EP3243844A4 is classified under aqueous emulsion polymerization with emulsifying agents.
- Both patents target styrene-content reduction, but US20190112458A1 discloses explicit butadiene:styrene feed ratios of 100:25 to 100:35, whereas EP3243844A4 provides only title/classification evidence for low combined styrene content without numerical recipe data in the extracted block.

- US20190112458A1 includes detailed temperatures of 45-55 °C for SBR synthesis and 0 °C for initiator formation; EP3243844A4 does not expose temperatures, pressures, initiators, or chain-transfer agents in the extracted evidence.
- US20190112458A1 uses post-polymerization acid treatment and ethanol precipitation, while EP3243844A4 does not disclose coagulation or workup steps in the extracted evidence.
- The two patents represent complementary routes to low-styrene SBR: a solution-polymerized hydroxyl-terminated SBR intermediate versus a direct low-combined-styrene emulsion SBR process.

4. CONCLUSION

The evidence supports two viable low-styrene SBR synthesis directions: explicit solution polymerization with controlled butadiene/styrene feed ratio, and aqueous emulsion polymerization aimed directly at low combined styrene content. The strongest actionable parameters come from US20190112458A1, where butadiene:styrene ratios of 100:25 to 100:35, synthesis temperatures of 45-55 °C, and controlled active-lithium ratios are disclosed. EP3243844A4 confirms that low combined styrene content is also pursued via emulsion polymerization, but the extracted evidence lacks detailed operating conditions. For low styrene SBR development, the best-supported synthesis strategy is to tightly control monomer feed ratio and polymerization route, with solution polymerization offering the most explicit recipe data in the provided corpus.

5. REFERENCES

- US20190112458A1 | Solution Polymerized Styrene-Butadiene Rubber-Polyurethane Elastomer Material ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US20190112458A1/en>
- EP3243844A4 | Method for preparing styrene-butadiene rubber having low combined styrene ... | LG Chem Ltd | EP | 2018 | <https://patents.google.com/patent/EP3243844A4/en>